# 01 Architecture Diagram

Deterministic core on the left, Gemini-calling services on the right. One Cloud Run service, backed by Firestore and Cloud Storage.

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PEOPLE

**Learner**
Voice and taps; Grade 4–12 demos

**Parent**
Reads the Parent View; "See why" on any claim

**Admin**
Uploads textbook + syllabus PDFs per board / grade / subject

voice taps
read
upload

BROWSER

**Web app**
React 19 + Vite + Tailwind
- Guided chalkboard + pictures
- Voice session (16 kHz PCM)
- Learner card, Tutor-reasoning panel
- Parent View ⇄ "Under the hood"
- Admin content library
- Maya / Leo profile picker (DEMO_MODE)

HTTPS /api/*
WSS /ws/live

**Recorded caveats**

- Model IDs (gemini-3.8-live, 3.6-flash…) are marked UNVERIFIED in AUDIT_2026-10-09 and the Cloud Run runbook; names here are as written in the docs.
- Firebase Hosting in front of Cloud Run is optional (TECHNICAL_SPEC §1a, ANANTA_CLOUD_RUN_PLAN).
- Spaced-review scheduling (T15) and EV-02 are not built.

CLOUD RUN service ananta · Node 22 · server.ts

DETERMINISTIC CORE · no GenAI

**Teaching Plan compiler**
Rules R-TARGET … R-TALK → plan; delta after each answer

plan

**Persona composer**
One fixed identity + hard rules H1–H13; plan block appended

system instruction

**Learner model engine**
BKT pKnown · evidence ladder · misconception ledger · Beta(α,β)

evidence

**Claim + report validators**
Claims need real event ids; parent sentences must cite evidence

gate

**Quality gates**
Lints, fact-checker, serve rules; failures are quarantined

gate

**AI gateway**
Every Gemini call: role → model, timeout, retry, fallback, structured log

SERVICES THAT CALL GEMINI

**REST API + session start**
Learners, curricula, session, parent-report, admin routes

**Live WebSocket proxy**
/ws/live ⇄ Gemini Live; server owns the chalkboard; tool dispatch

audio ⇄ tool calls

**Reasoning assessor**
Classifies the child's reasoning against the misconception catalogue

answer → JSON verdict

**Profiler + parent letter**
Session end: scoped claims. Parent View: letter, tip

evidence → claims, letter

**Content pipeline**
PDF → curriculum → lessons → pictures → photo → critic

GOOGLE AI

**Gemini 3.8 Live** (GOOGLE)
Voice, Puck voice, VAD; 8 function-calling tools in guided mode

**Gemini 3.6 Flash** (GOOGLE)
Fast diagnosis, 6.5 s budget

timeout

**Gemini 3.1 Flash Lite** (GOOGLE)
Fallback for diagnosis and lessons

**Gemini 3.8 Flash** (GOOGLE)
PDF ingest, lessons, profiler, parent letter, critic reviews

**Gemini Vision** (GOOGLE)
Critic for generated photos

**Gemini Image generation** (GOOGLE)
Interest-themed lesson photo

**Google Files API** (GOOGLE)
PDF chunks (60 pages, 40 MB)

Admin SDK server-only
photo upload
read at start

GOOGLE CLOUD · DATA, SECRETS, IDENTITY

**Firestore** (GOOGLE)
learners + events/plans/sessions · curricula · pregen · guidedScripts

**Cloud Storage** (GOOGLE)
pregen-images/ verified photos (public read, that prefix only)

**Secret Manager** (GOOGLE)
GEMINI_API_KEY
ADMIN_TOKEN

**Firebase Auth** (GOOGLE)
Configured; bypassed when DEMO_MODE=true

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: README.md (Architecture; Google AI & Cloud Services) · docs/ARCHITECTURE.md §1–7 · docs/TECHNICAL_SPEC.md §1a, §3 · docs/HACKATHON_AUDIT.md · docs/ADAPTIVE_TEACHING.md · docs/ANANTA_CLOUD_RUN_PLAN.md

# 02 Technical Diagram

Modules, APIs, tool calls and data flows. Numbered badges mark every point where Ananta touches a Google service.

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## CLIENT

**Guided chalkboard**
GuidedBoard.tsx · useGuidedTutor.ts · TutorModeSwitch (pace)

**Board pictures**
BoardVisualView (SVG) · Board3DView (Three.js) · ScenePanel

**Learner + reasoning views**
LearnerProfilePanel · TutorReasoningPanel · Onboarding · SubjectSelector

**Parent**
ParentHome → ParentReport (default) ⇄ ParentPortal ("Under the hood")

**Admin**
LoginScreen link → AdminLibrary · CurriculumUpload

**Live transport**
utils/liveWs.ts: same-origin WSS, 16 kHz PCM up, audio down

## server.ts Express REST + ws /ws/live + static dist/

### REST /api/*

**Learners & session**
GET/POST/DELETE /learners[/:id]
POST /learners/:id/onboarding
POST /learners/:id/tutor-name
POST .../misconceptions/:id/dispute
GET /learners/:id/events
POST /session/start
POST /session/:id/assess · /end
GET /learner-context/:id/:subject
POST /tutor/turn (text channel)

**Parent & content**
GET /learners/:id/parent-report
POST .../parent-report/generate
POST /generate-lesson
POST /update-diagram
POST /generate-image
GET /curricula · /catalog
GET /guided-script · /tutor-mode
GET /health

**Admin (requireAdmin)**
POST /curriculum/upload
GET /curriculum/jobs/:id
GET /admin/check
GET/DELETE /admin/courses[/:id]

Guards: requireAuth + requireOwnership on learner routes (bypassed when DEMO_MODE=true) 10

### WEBSOCKET /ws/live 1

client → server: audio · text · tool_response · close
server → client: guided_state · learner_update_v2 · plan_update · tutor_reasoning

**Gemini tool calls the server acts on**

- assess_child_reasoning → assessor → recordReasoningEvidence → plan delta (non-blocking, never held past the voice turn)
- record_confusion_signal → affect counters
- update_diagram → store read ≤ 1.2 s, answers with step names; never generates
- Guided: advance_beat · next_line · fill_slot · write_aside · show_picture · resume_lesson · skip_to_practice · teach_skipped
- All other board tools are pure relays and never touch the learner model

## SESSION PIPELINE deterministic steps in amber, model calls in green

**Session start**
getRepo → profile → compileTeachingPlan (R-TARGET ... R-TALK)

**Compose prompt**
renderPlanForPrompt + persona + board pictures + curriculum → one system instruction

**Per answer** 2
assessReasoning (JSON) → ledger · BKT · ladder · Beta strategy → compilePlanDelta

**Session end** 3
runProfiler → claimValidator → saveProfile + SessionSummary

**Repository (getRepo)** 7
K_SERVICE or USE_FIRESTORE_LEARNERS → FirestoreLearnerRepository (Admin SDK); otherwise FileLearnerRepository (data/*.json). Change both together.

**Content generators** 3 4 5 6
lessonGen · board visuals · photoGen · critic · guided scripts: draft → lint → critic → repair → quarantine (fail closed)

**AI gateway src/ai/gateway.ts** 1 2 3 4 5 9
Roles fast · strong · review · live · image (MODEL_FAST / STRONG / REVIEW / LIVE / IMAGE) · generateJSON(schema) · timeout · transient retry · model fallback · structured log {call, model, ms, ok}

## PERSISTENCE (Firestore collections / local files)

**learners/{id}** 7
profile · conceptStates · strategyProfile · claims · parentNarratives

**.../events · plans · sessions** 7
append-only evidence, plan versions, session summaries

**curricula · pregen** 8 7
courses; lessons, board pictures (as JSON strings), quality record

**guidedScripts** 7
runtime lessons; re-read by server.ts every 10 s

## GOOGLE INTEGRATION POINTS

1 **Gemini 3.8 Live**
Live WebSocket; system instruction + tools; Puck voice; VAD

2 **Gemini 3.6 Flash (fast)**
assessReasoning: JSON schema, 6.5 s budget; cascade to 3.1 Flash Lite

3 **Gemini 3.8 Flash (strong)**
Profiler (20 s) · parent narrative (25 s) · PDF extract · lessons · guided scripts

4 **review role + Gemini Vision**
Independent critic; photo review (inline image)

5 **Gemini Image generation**
Scene planned from key facts; no text in image

6 **Google Files API**
pdfIngest: 60-page / 40 MB chunks

7 **Firestore (Admin SDK)**
Client rules: allow read, write: if false

8 **Cloud Storage**
pregen-images/ public read

9 **Secret Manager**
GEMINI_API_KEY · ADMIN_TOKEN

10 **Firebase Auth**
ID-token check in requireAuth; off in DEMO_MODE

Numbered badges inside the server mark the call site of each Google service listed above (same number).

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/ARCHITECTURE.md §4–7 · docs/TECHNICAL_SPEC.md §3–6 · docs/GUIDED_TUTOR.md §5, §13 · docs/ADAPTIVE_TEACHING.md · docs/PARENT_VIEW.md §3–4 · docs/CURRICULUM.md §4, §8 · README.md

# 03 Functional Diagram

What the learner and parent experience, and which functions produce it: persona, learner model, per-child teaching plan.

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**SESSION ARC (what the child goes through)**

OPEN → PREREQ → TEACH → CHECK → ELICIT → DIAGNOSE → REMEDIATE → APPLY → TRANSFER → TEACH-BACK → CLOSE

Child can say “I don’t know”, “explain it differently”, “I think you’re wrong”; the tutor responds without shaming and changes representation, not just wording.

**Learner**
Grade 4 demo: Maya, Leo (Simulated)

**Parent**
reads the Parent View

### Tutor persona
Fixed identity composed by code; same character every session.
Hard rules H1–H13 are always in the system instruction.
Voice: Gemini Live (Puck). Board: guided chalkboard with pictures, asides, fill-in slots.
Can pause the lesson to answer, then resume_lesson.

speaks

voice + tools

### Gemini Live
Speaks and listens; calls guided-board tools and assess_child_reasoning.

### Learner model (deterministic)
Bayesian knowledge tracing: provisional mastery needs pKnown ≥ 0.80 and ≥ 2 distinct level-3+ items.
Evidence ladder L0–L5: levels of demonstrated understanding.
Misconception ledger: suspected → confirmed → resolved.
Strategy profile: Beta(α,β) per concept type × representation; avoid after ≥ 3 uses with mean < 0.3; decays 0.9 per 30 days.

compile

### Per-child teaching plan
Compiled from the learner model before every session by rules R-REVIEW, R-TARGET, R-PREREQ-FIRST, R-PROBE, R-WATCH, R-REP, R-AVOID, R-SCAFFOLD, R-DIFF, R-FAST, R-PACE, R-GOAL, R-TALK.
Updated mid-session by compilePlanDelta.

assessment → evidence

### Gemini 3.6 Flash (assessor) · Gemini 3.8 Flash (profiler)
Read the child’s explanation; write a structured assessment and the end-of-session profile.

profile / letter

### Validators (code)
Claim validator and parent-report validator reject unsupported statements before anything is saved or shown.

validated

### Parent View
Plan card first (what the next sessions will do and why).
“See why” replays the evidence.
No scores or jargon; the parent can dispute a recorded misconception.
Demo: Leo’s mix-up “−5 > −1” is shown with its evidence.

reads

Teacher: not modelled as a user in the docs. The product is positioned as teacher + AI companion; classroom teaching stays with the teacher.

Out of scope for this view: content ingest (diagram 05c) and deployment (diagram 08).

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/TUTOR_PERSONA.md · docs/LEARNER_MODEL.md · docs/TEACHING_PLAN.md · docs/ADAPTIVE_TEACHING.md · docs/PARENT_VIEW.md · docs/GUIDED_TUTOR.md

04

# Use-Case Diagram

Who uses Ananta and what for. Scenarios come from the session arc, guided-tutor tools, Parent View and content pipeline in the docs.

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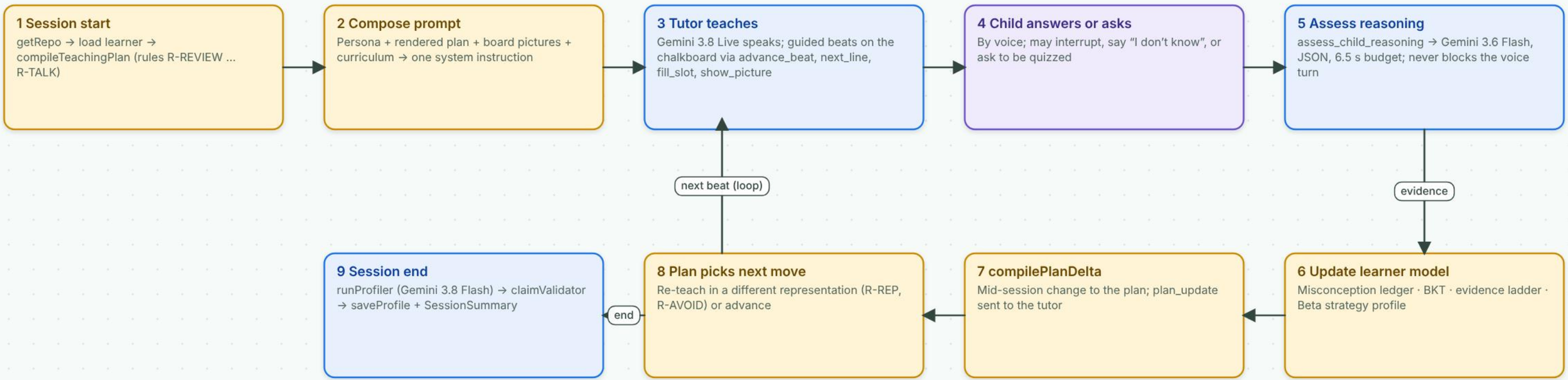
Sources: docs/FUNCTIONAL_SPEC.md · docs/GUIDED_TUTOR.md · docs/PARENT_VIEW.md · docs/CURRICULUM.md · docs/ADAPTIVE_TEACHING.md · docs/TECHNICAL_SPEC.md

05a

# Process: Learning Adaptation Loop

One session, step by step. Amber = deterministic code, blue = Google model call, violet = the child.

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### What the loop guarantees (from the docs)

- A correct answer alone does not advance the child: reasoning is assessed (memorised vs understood vs transfer).
- A misconception moves suspected → confirmed → resolved only on repeated evidence, never on one ambiguous answer.
- A representation that fails (≥ 3 uses, mean < 0.3) is avoided; the strategy profile decays 0.9 per 30 days.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

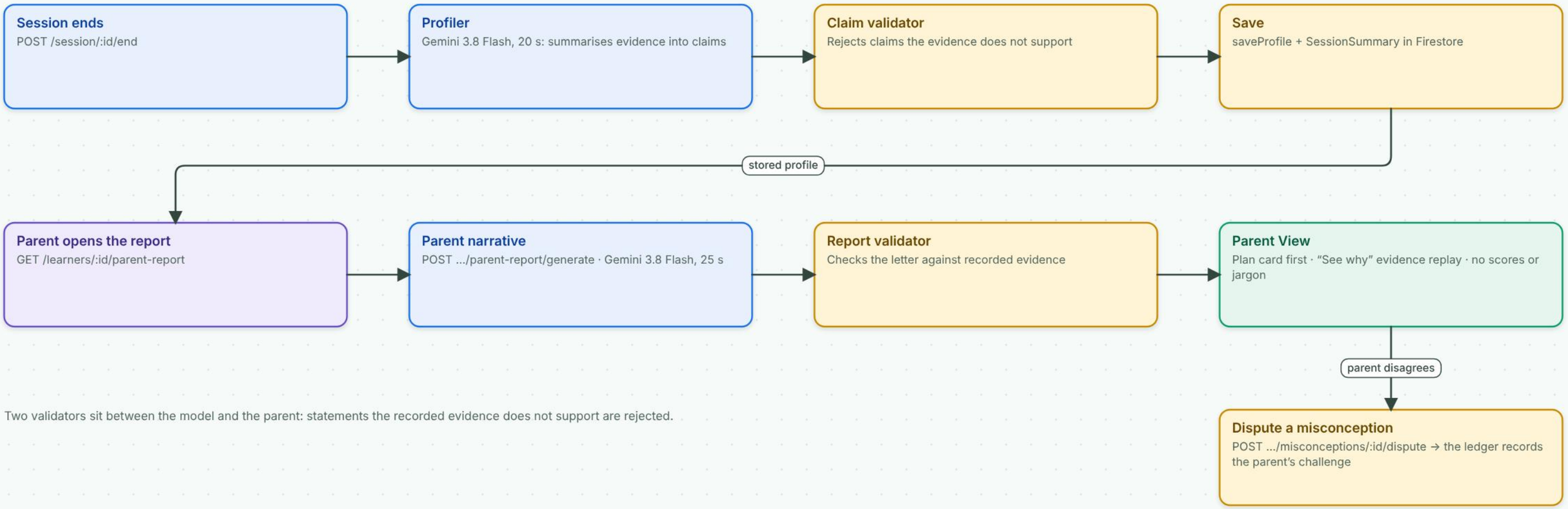
Sources: docs/ADAPTIVE_TEACHING.md · docs/TEACHING_PLAN.md · docs/LEARNER_MODEL.md · docs/GUIDED_TUTOR.md §5 · docs/TECHNICAL_SPEC.md

05b

# Process: Parent View

From the end of a session to what the parent reads, with the checks in between. Illustrative learners are Maya and Leo (Simulated).

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Two validators sit between the model and the parent: statements the recorded evidence does not support are rejected.

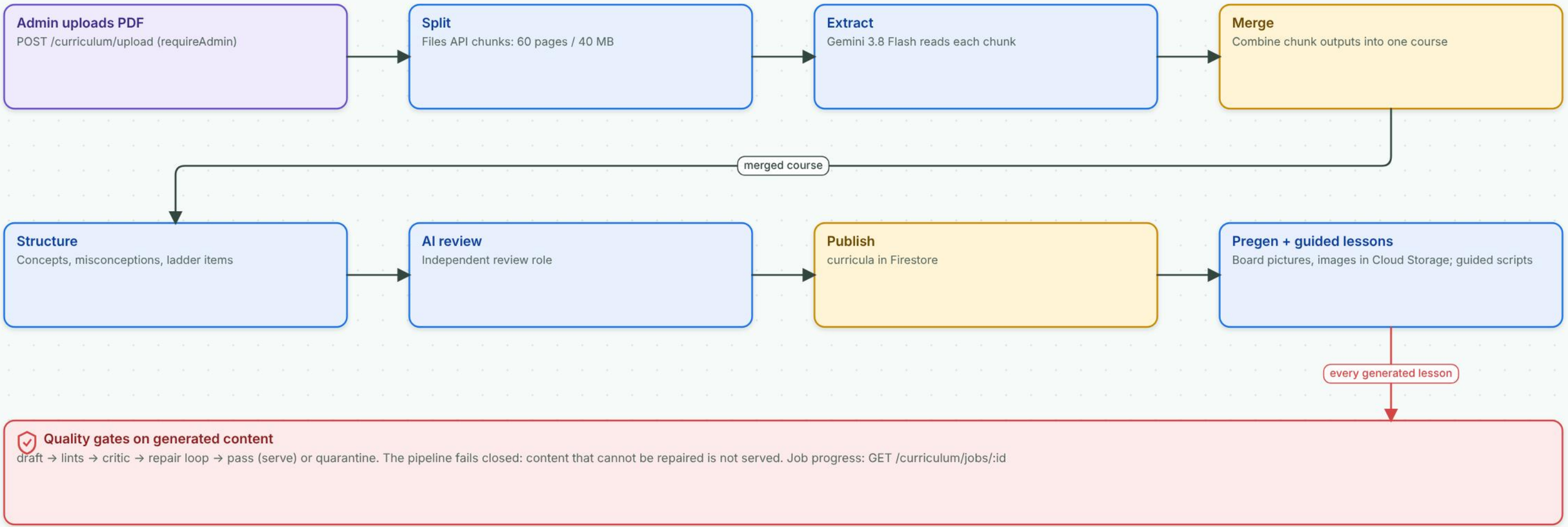
Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/PARENT_VIEW.md · docs/LEARNER_MODEL.md · docs/TECHNICAL_SPEC.md · docs/ARCHITECTURE.md

# 05c Process: Curriculum Ingest and Lesson Quality

Admin uploads a PDF; the course passes quality gates before a child can see it. Failures are quarantined (fail closed).

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Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

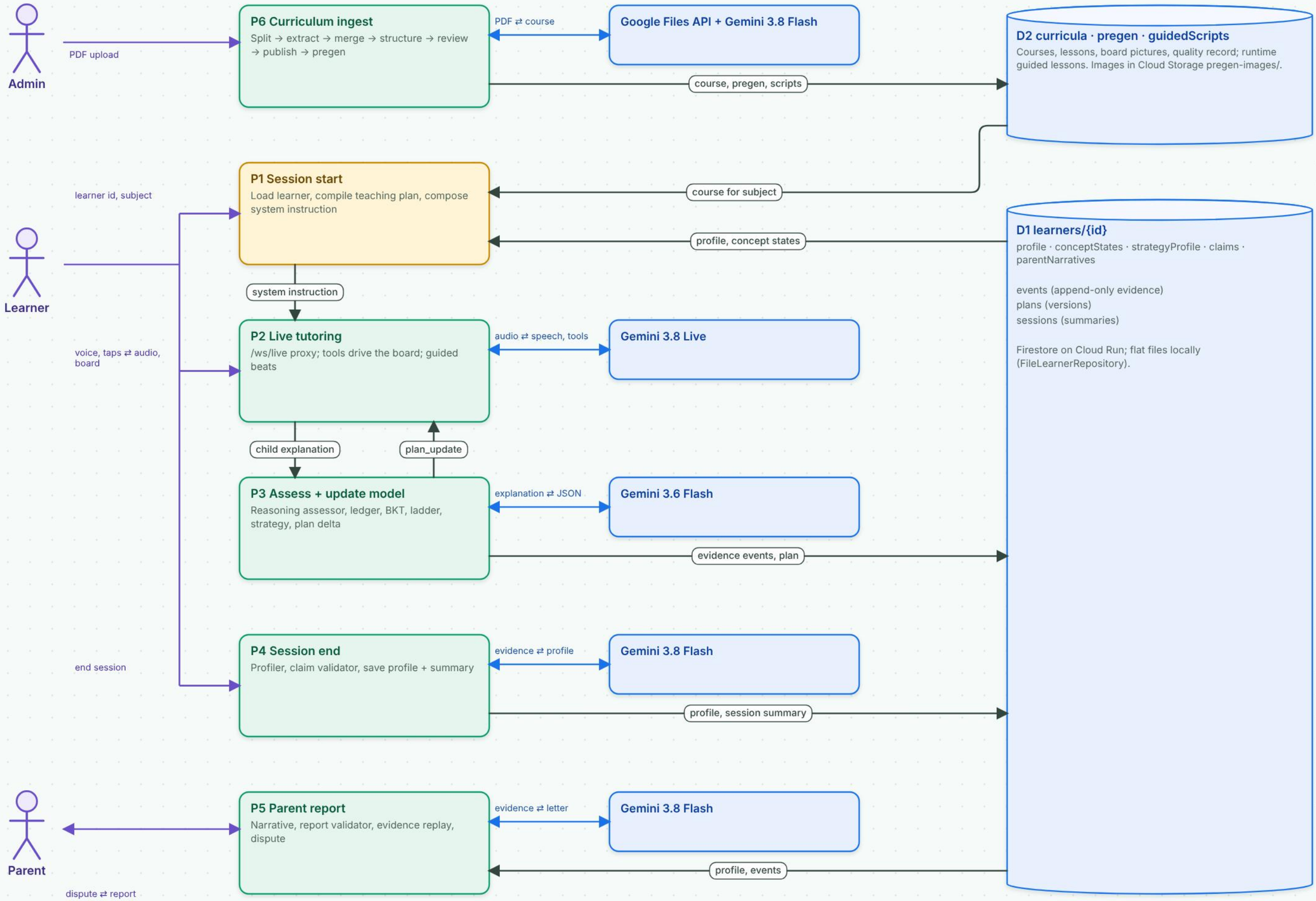
Sources: docs/CURRICULUM.md §4, §8 · docs/CH1_CONTENT_SPEC.md · docs/CH1_FIX_PLAN.md · docs/TECHNICAL_SPEC.md · docs/GUIDED_TUTOR.md

06

# Data Flow Diagram (Level 1)

What data moves between people, processes, Google models and stores. Processes = rounded boxes, stores = cylinders.

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Note: P2 also reads the guided lesson script from D2 (re-read every 10 s); the arrow is omitted for clarity.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

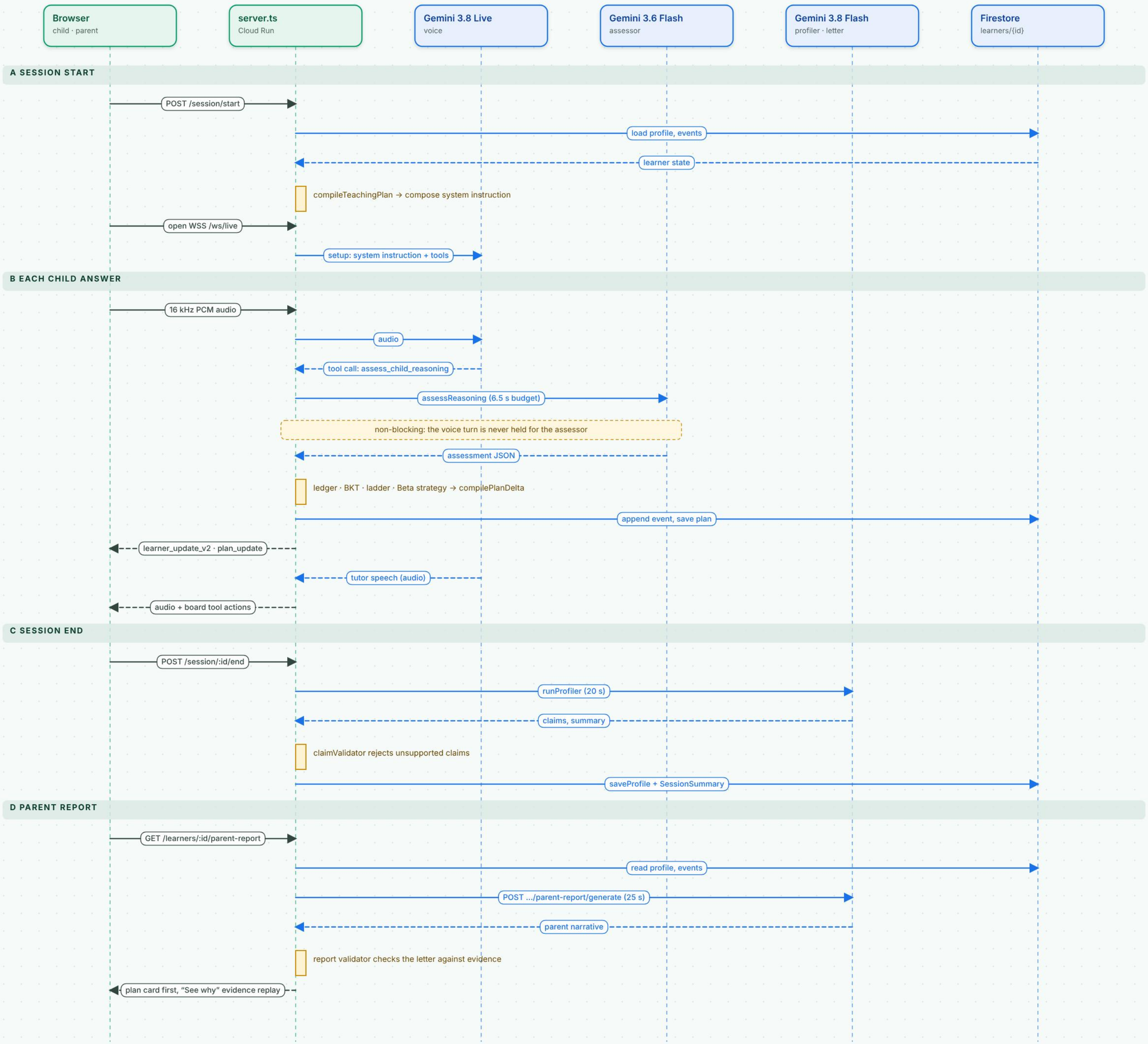
Sources: docs/ARCHITECTURE.md §4–7 · docs/TECHNICAL_SPEC.md §3–6 · docs/LEARNER_MODEL.md · docs/PARENT_VIEW.md · docs/CURRICULUM.md

07

# Sequence Diagram

One session end to end: start, a child's answer, session end, then the parent report. Blue lifelines are Google services.

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Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/ARCHITECTURE.md §5–6 · docs/TECHNICAL_SPEC.md §3 · docs/ADAPTIVE_TEACHING.md · docs/GUIDED_TUTOR.md §5 · docs/PARENT_VIEW.md §3–4

# 08 Deployment Diagram

One Cloud Run service carries the API, the Live WebSocket proxy and the built web app. Data and secrets live in Google Cloud.

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**Firebase Hosting**
Optional; not required

**Browser**
Learner · Parent · Admin

HTTPS · WSS /ws/live

**Cloud Run service ananta**

**Container node:22-bookworm-slim**
server.ts: REST /api/*, WebSocket /ws/live, built Vite SPA (React 19, Tailwind) from dist/. Listens on port 8080.

**Service settings**
Session affinity on (WebSocket)
Request timeout 3600 s
Min 0 / max 3 instances
1 vCPU / 1 GiB, no-cpu-throttling

**Runtime service account ananta-run**
roles/datastore.user · secretmanager.secretAccessor · storage.objectAdmin

model calls → **Gemini API (Google AI)**
Live, 3.6 Flash, 3.8 Flash, 3.1 Flash Lite, Vision, Image, Files API

Admin SDK → **Firestore**
learners, curricula, pregen, guidedScripts. Client rules deny all reads and writes.

images → **Cloud Storage**
pregen-images/ (public read)

env at deploy → **Secret Manager**
GEMINI_API_KEY · ADMIN_TOKEN

verify token → **Firebase Auth**
ID-token check; bypassed when DEMO_MODE=true

deploy

**Source repo**
pythagoras-tutor

**Cloud Build**
builds the container image

**Artifact Registry**
stores the image

**Open point**
The region is not stated consistently across the deployment docs. Confirm it before submission.

Build and deploy steps are listed in docs/ANANTA_CLOUD_RUN_PLAN.md; this diagram shows only the resulting nodes.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

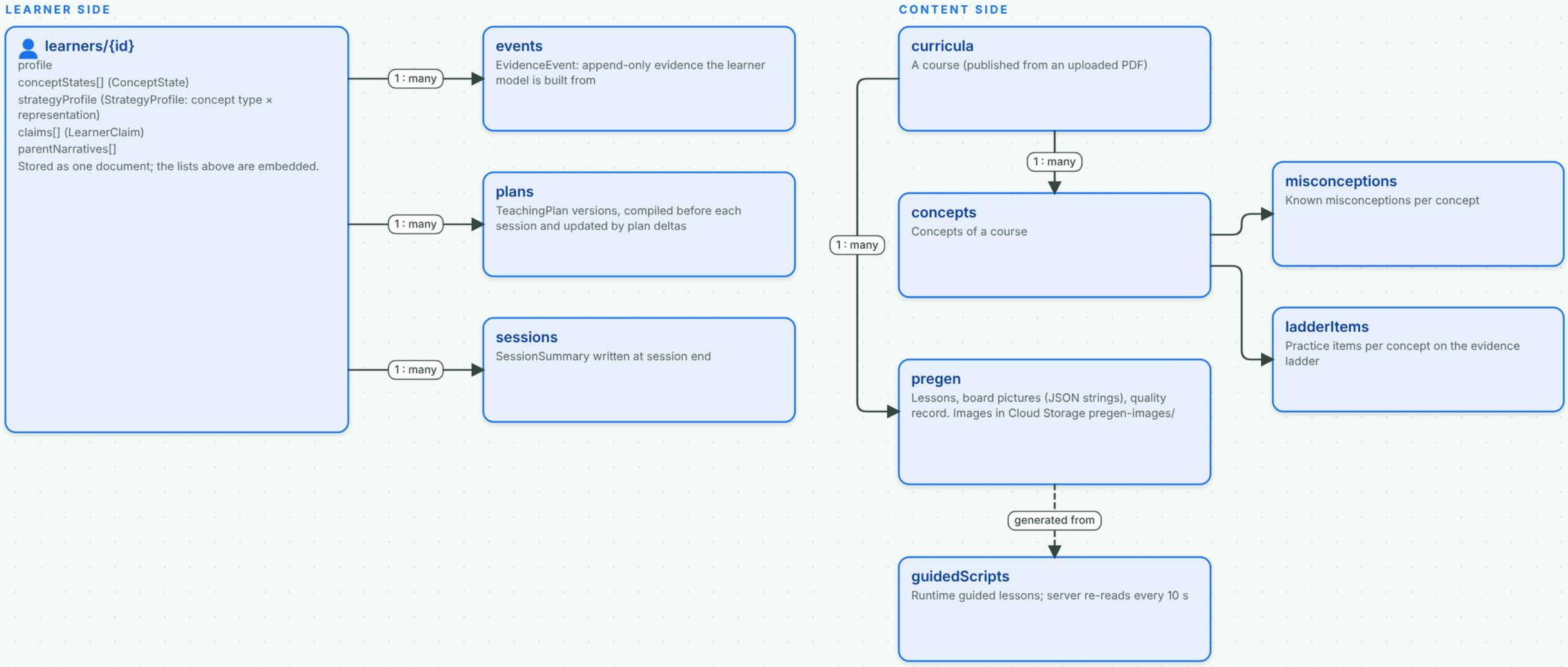
Sources: docs/DEPLOYMENT_PLAN.md · docs/ANANTA_CLOUD_RUN_PLAN.md · docs/ANANTA_MIGRATION_PLAN.md · docs/ANANTA_BEGINNER_GUIDE.md · docs/ARCHITECTURE.md §7

09

# Entity-Relationship Diagram

Firestore collections and what each holds. Only fields and links the docs name are drawn; the dashed link is derived.

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Not drawn: field-level schemas (see docs/TECHNICAL_SPEC.md) and the local flat-file store (data/*.json) that mirrors learners when Firestore is not configured.

Access: server only (Admin SDK). Client security rules: allow read, write: if false.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/TECHNICAL_SPEC.md (schema types) · docs/LEARNER_MODEL.md · docs/CURRICULUM.md · docs/ARCHITECTURE.md §7 · docs/ADAPTIVE_TEACHING.md

10

# Component Diagram

Code components grouped by layer, with the Google service each layer depends on.

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## CLIENT (React 19 · Vite · Tailwind · Three.js)

**Guided board**
GuidedBoard · useGuidedTutor · TutorModeSwitch

**Board pictures**
BoardVisualView · Board3DView · ScenePanel

**Learner views**
LearnerProfilePanel · TutorReasoningPanel · Onboarding · SubjectSelector

**Parent**
ParentHome · ParentReport · ParentPortal

**Admin**
LoginScreen · AdminLibrary · CurriculumUpload

**Live transport**
liveWs.ts (WSS, PCM audio)

HTTPS

WSS

## SERVER EDGE (server.ts on Cloud Run)

**REST routes /api/***
Learners, session, parent report, curriculum, health

**/ws/live proxy**
Relays audio and tools to Gemini Live; sends learner updates

**Guards**
requireAuth · requireOwnership · requireAdmin (bypassed in DEMO_MODE)

**Static host**
Serves the built SPA from dist/

compile plan, validate

## DOMAIN (deterministic, no GenAI)

**Teaching plan compiler**
Rules R-REVIEW … R-TALK; compilePlanDelta

**Persona composer**
Fixed identity, hard rules H1–H13

**Learner model engine**
BKT · evidence ladder · misconception ledger · Beta strategy

**Validators**
Claim validator · parent-report validator

**Quality lints / serve rules**
Gate content before it is served

evidence

## AI SERVICES (call Gemini through the gateway)

**Reasoning assessor**
assessReasoning → JSON

**Profiler + parent letter**
runProfiler · parent narrative

**Content pipeline**
pdfIngest · structure · lessonGen · guided scripts

**Board visuals + photo review**
board visuals · photoGen · photo critic

**Critic + repair loop**
Independent review; quarantine on failure

## AI GATEWAY

**src/ai/gateway.ts**
Roles fast · strong · review · live · image (MODEL_* env) · generateJSON(schema) · timeout, transient retry, model fallback, structured logs

model calls

## PERSISTENCE

when K_SERVICE is set

**Repository getRepo()**
K_SERVICE or USE_FIRESTORE_LEARNERS → Firestore; otherwise flat files

**FileLearnerRepository**
data/*.json (local)

**FirestoreLearnerRepository**
Admin SDK

**Google Cloud + AI**
Gemini API · Firestore · Cloud Storage · Secret Manager · Firebase Auth

Rule from docs: when the repository interface changes, change the Firestore and file implementations together.

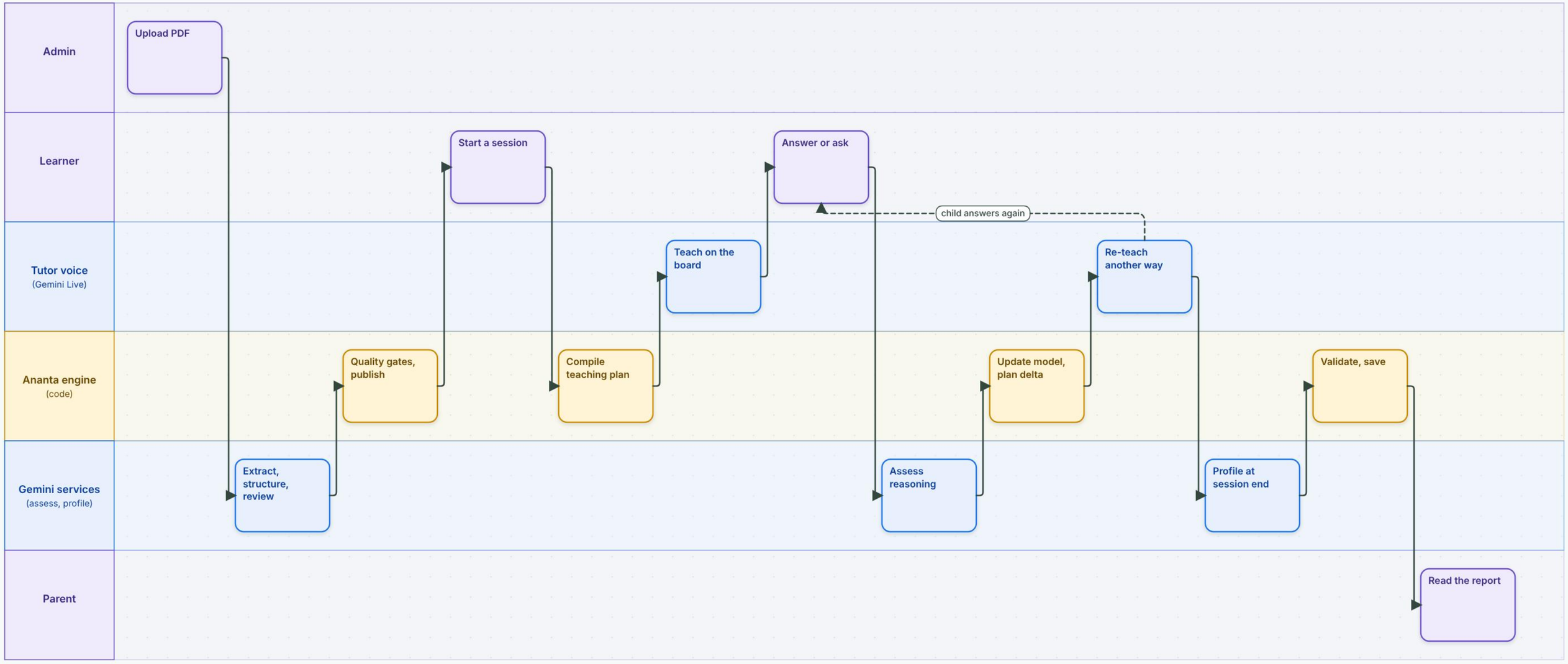
Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/ARCHITECTURE.md §4–7 · docs/TECHNICAL_SPEC.md §3 · docs/GUIDED_TUTOR.md §13 · docs/ADAPTIVE_TEACHING.md · docs/CURRICULUM.md §4 · README.md

# 11 Workflow Swimlane Diagram

Who does what, in time order, from curriculum upload to the parent's report. Lanes: Admin, Learner, Tutor voice, Ananta engine (code), Gemini services, Parent.

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Parent can also dispute a recorded misconception; the ledger records the challenge. Steps are ordered by time, not drawn to scale.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/CURRICULUM.md · docs/GUIDED_TUTOR.md · docs/ADAPTIVE_TEACHING.md · docs/LEARNER_MODEL.md · docs/PARENT_VIEW.md · docs/ARCHITECTURE.md

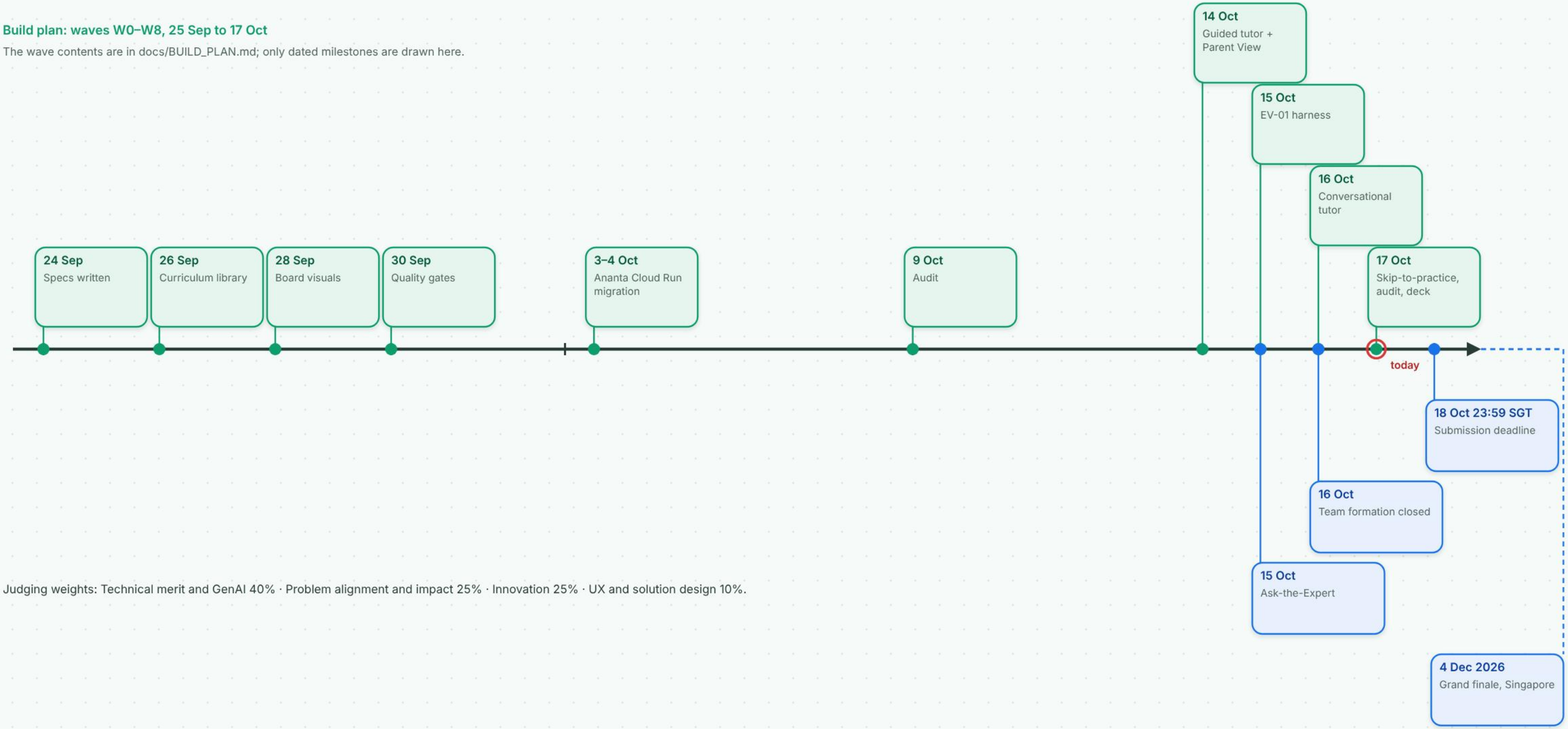
# Timeline and Roadmap

Build milestones (above the line) and hackathon dates (below). The deadline is the next day.



## Build plan: waves W0–W8, 25 Sep to 17 Oct

The wave contents are in docs/BUILD_PLAN.md; only dated milestones are drawn here.




Judging weights: Technical merit and GenAI 40% · Problem alignment and impact 25% · Innovation 25% · UX and solution design 10%.

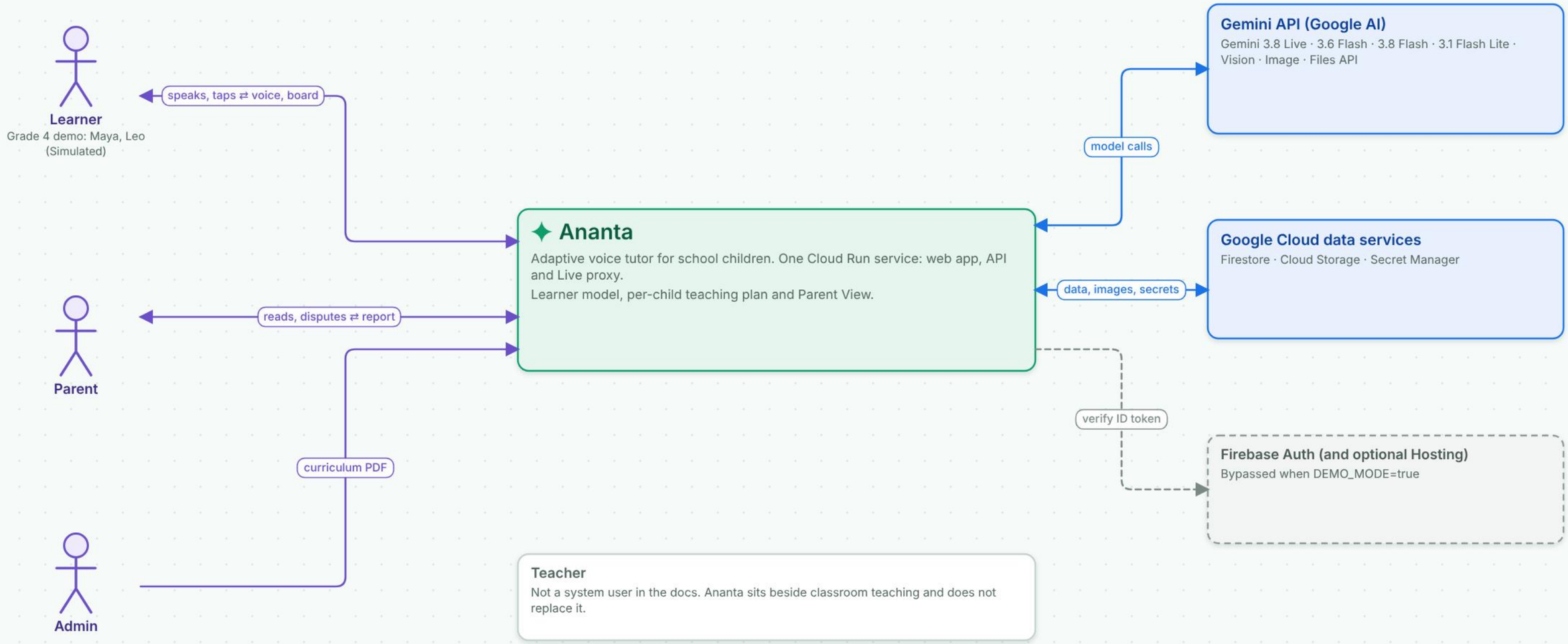
Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/BUILD_PLAN.md · docs/AUDIT_2026-10-09.md · docs/ANANTA_MIGRATION_PLAN.md · docs/GUIDED_TUTOR.md · hackathon dates from the AI Builder Cup pages recorded in the project

# System Context Diagram

Ananta as one system: the people around it and the Google services it depends on.






Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/ARCHITECTURE.md §1 · docs/FUNCTIONAL_SPEC.md · docs/PARENT_VIEW.md · docs/CURRICULUM.md · README.md

14

# Ananta at a Glance

One page for judges: the problem, the system, the Google tools it uses, and where each judging criterion is evidenced.

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## THE PROBLEM AND THE IDEA

**The problem**
Children often meet ideas they do not understand but hesitate to ask: embarrassment, peer judgment, not knowing what to ask. A teacher cannot give every child unlimited one-to-one explanation.

**Ananta**
A voice-first adaptive tutor. The child can say “I don’t know” or “explain it differently” without judgment. It checks understanding, not just answers, and changes how it teaches when something does not land.

**Teacher + AI companion**
Not a replacement for the teacher. Gives private questioning, formative checks, misconception detection and a parent-readable report between classes.

## THE LEARNING LOOP

**1 Plan**
Teaching plan compiled from the learner model

→ **2 Teach**
Gemini Live speaks; chalkboard beats and pictures

→ **3 Assess**
Child’s reasoning read by Gemini 3.6 Flash

→ **4 Update**
Misconception ledger, BKT, strategy profile

→ **5 Adapt**
Plan delta: change representation or advance

→ **6 Report**
Profiler and parent letter, validated

## HOW IT IS BUILT (green = Ananta code · amber = deterministic · blue = Google)

**Browser**
React 19 + Vite SPA
Guided chalkboard, voice, Parent View, Admin

↔

**Cloud Run ananta** one Node/Express service

- **Deterministic core**
  Plan compiler · persona · learner model · validators · quality gates
- **Gemini-calling services**
  Assessor · profiler · parent letter · lessons · critic

↔

**Google AI + Cloud**
Gemini 3.8 Live · 3.6 Flash · 3.8 Flash · 3.1 Flash Lite · Vision · Image · Files API
Cloud Run · Firestore · Cloud Storage · Secret Manager · Firebase Auth

## WHO USES IT

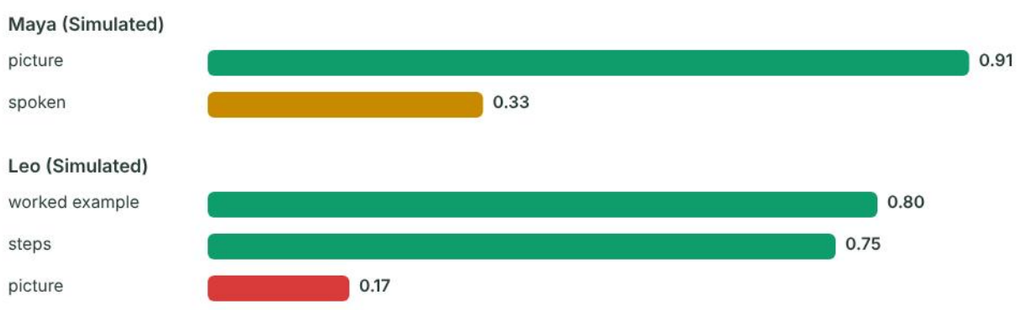
**Learner**
Learns by voice with the board
Interrupt, ask, “quiz me”
Misconceptions caught and re-taught

**Parent**
Plan card first, “See why” evidence replay
No scores or jargon
Can dispute a recorded misconception

**Admin**
Uploads a curriculum PDF
Extract, structure, AI review, publish
Quality gates fail closed

## EVIDENCE OF ADAPTATION (demo learners, Simulated)

**Strategy profile: mean success per representation (Beta α,β)**




Maya does better with pictures; Leo with worked examples and steps, so the avoid rule (R-AVOID) can retire pictures for him. Leo’s recorded mix-up: “−5 > −1”.

**Judging criteria and where the project evidences them**

| | | |
|---|---|---|
| 40% | **Technical merit and GenAI** | Five model roles behind one gateway; Live voice with tool calls; deterministic learner model; fail-closed quality gates |
| 25% | **Problem alignment and impact** | Judgment-free asking; parent report with evidence; teacher + AI stance |
| 25% | **Innovation and creativity** | Strategy changes per child per concept type; misconception ledger; assess without blocking voice |
| 10% | **UX and solution design** | Guided chalkboard, plan-first Parent View, no scores or jargon |

**Open points recorded in the docs**
Evaluation (EV-01): the docs disagree on whether results are reported or still pending. Confirm before quoting numbers. Region for Cloud Run is not stated consistently. Tool counts and the diagnosis time budget differ between docs. Model IDs for the newest Gemini names are not verified in the docs.

Weights are the published judging criteria recorded in the project. Scores shown are simulated learners used for the demo, not real children.

Ananta app code · Google tool · Deterministic (no GenAI) · Actor / user · Planned · optional · not built

Sources: docs/FUNCTIONAL_SPEC.md · docs/ARCHITECTURE.md · docs/ADAPTIVE_TEACHING.md · docs/LEARNER_MODEL.md · docs/PARENT_VIEW.md · docs/TRACEABILITY.md · docs/AUDIT_2026-10-09.md · hackathon criteria as recorded in the project

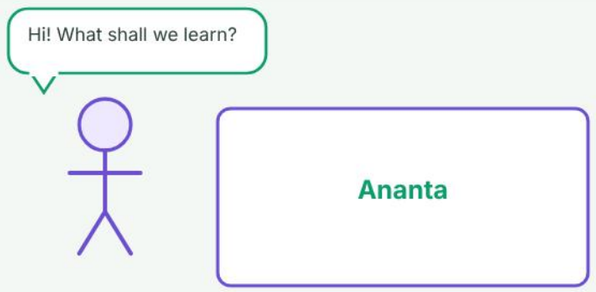
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# Storyboard: Leo's Session, then His Parent's View

Six panels built from documented behaviour (Leo, Grade 4, Simulated). The exact demo script may differ.

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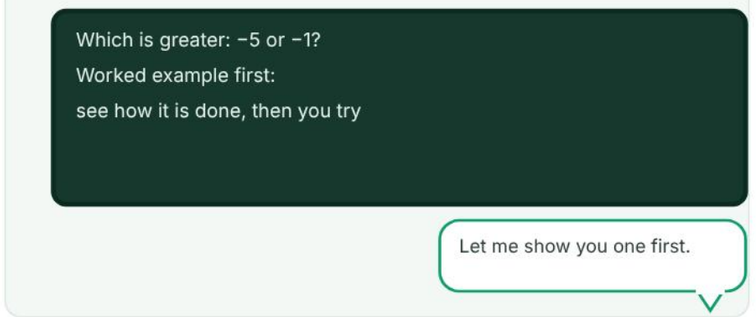
## 1 Leo starts a session




Leo opens Ananta and begins a lesson. No one else is watching.

**Behind the scenes**
Plan compiled from his stored profile before the first word is spoken (getRepo, compileTeachingPlan).

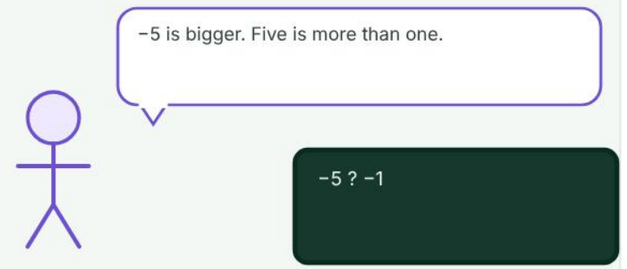
## 2 The tutor teaches




The tutor speaks and writes on the chalkboard. For Leo, a worked example comes first.

**Behind the scenes**
His profile favours worked examples (0.80) and steps (0.75) over pictures (0.17), so the plan picks the representation. Gemini 3.8 Live speaks.

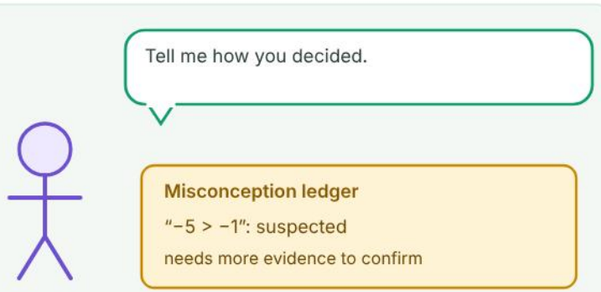
## 3 Leo gives an answer




Leo says −5 is bigger because five is more than one. He is not embarrassed to guess.

**Behind the scenes**
assess_child_reasoning reads the explanation, not just the answer. Gemini 3.6 Flash, JSON result, never blocks the voice.

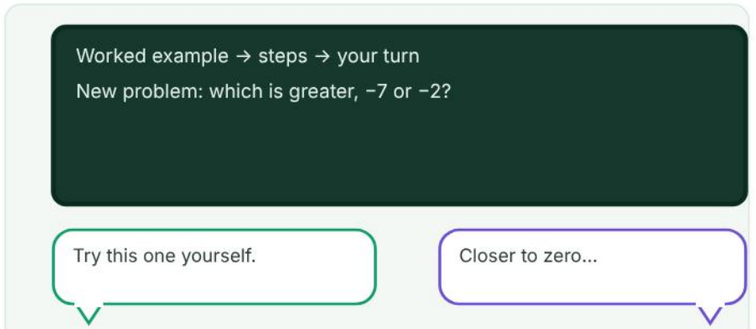
## 4 The tutor asks why




The tutor asks Leo to explain how he decided. A single answer does not label a misconception.

**Behind the scenes**
The ledger records the mix-up as suspected. Confirmed only with repeated evidence; the plan is updated (compilePlanDelta).

## 5 Teach it another way, then re-test




The tutor changes the representation and gives Leo a fresh problem to try himself.

**Behind the scenes**
Strategy profile and R-AVOID steer away from pictures for Leo. Transfer item tests whether he can apply it.

## 6 His parent reads the report




The plan card comes first. "See why" shows the evidence behind each statement.

**Behind the scenes**
Profiler, claim validator and report validator run first; unsupported statements are rejected. Parent can dispute.

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Sources: docs/ADAPTIVE_TEACHING.md · docs/LEARNER_MODEL.md · docs/PARENT_VIEW.md · docs/GUIDED_TUTOR.md · docs/TEACHING_PLAN.md

# Dashboard Mockups

Left: the Parent View (plan first, no scores). Right: the learner “under the hood” panel. Layouts are illustrative; numbers are the documented demo values for Maya and Leo (Simulated).



Parent View · Leo (Simulated), Grade 4

## Leo’s week

Written from what Leo actually did. No scores.

**PLAN · what happens next and why**

**Next sessions start with a worked example, then steps, then a fresh problem for Leo to try himself.**

Why: worked examples and steps have helped Leo on this kind of idea; pictures have not.

See why

**WHAT WE NOTICED**

**Leo may be mixing up which negative number is greater (for example −5 and −1).**

suspected  See why  I disagree

More evidence is needed before this is treated as confirmed.

**A SHORT LETTER**

Checked against the recorded evidence before you see it.

**SEE WHY · evidence replay**

- Leo’s explanation of −5 and −1
- The tutor’s follow-up question
- Result of the new problem

Under the hood · learner panel (illustrative)

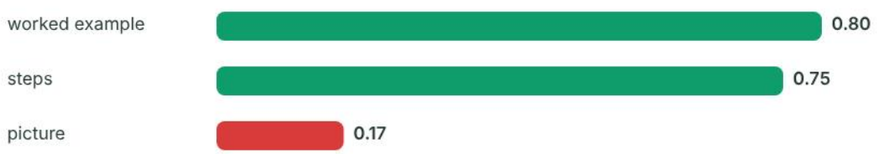
## What the tutor has learned about each child

Strategy profile: mean success per representation (Beta α,β)

**Maya (Simulated)**




**Leo (Simulated)**




**Misconception ledger**

“−5 > −1” Leo  suspected

Moves suspected → confirmed → resolved only on repeated evidence.

Mastery: provisional when pKnown ≥ 0.80 with ≥ 2 distinct level-3+ items.

**Not shown to parents**

Strategy numbers and the ledger live here; the Parent View stays plain-language with no scores or jargon.

Ananta app code  Google tool  Deterministic (no GenAI)  Actor / user  Planned · optional · not built

Sources: docs/PARENT_VIEW.md · docs/LEARNER_MODEL.md · docs/ADAPTIVE_TEACHING.md · docs/TECHNICAL_SPEC.md (LearnerProfilePanel, ParentReport, ParentPortal)